Class 5: Data Viz with ggplot2

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Background

There are many graphics systems in R for making plots and figures. These include so-called *"base R" graphics* like the `plot()` function and add on packages like **ggplot2**.

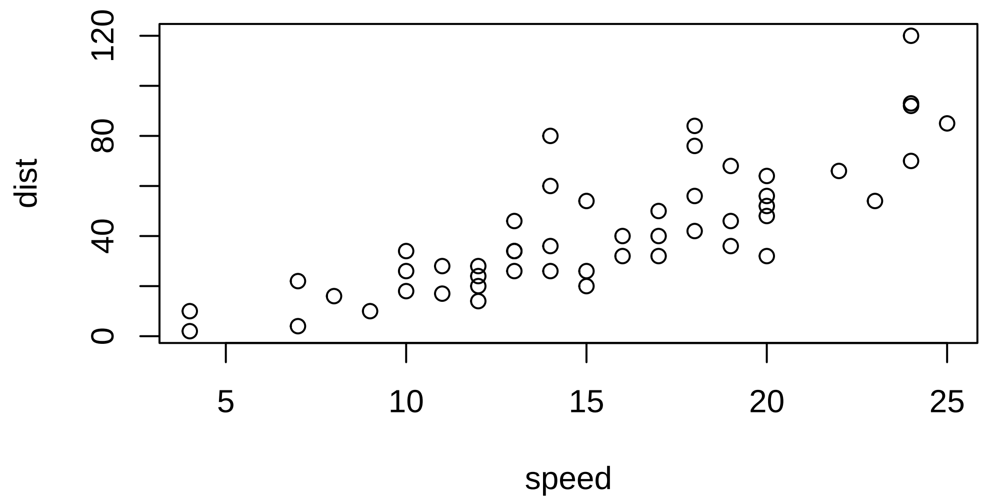
Let's compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```r
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

```r
plot(cars)
```



Before I can use ggplot2 I need to install it on my computer. To do this, we can use the function `install.packages("ggplot2")`

> *NOTE* We never run `install.packages()` in our quarto doc(We run it once only in R console) as it re-installs package every time we render.

Once installed we need to load up the package into our R brain :

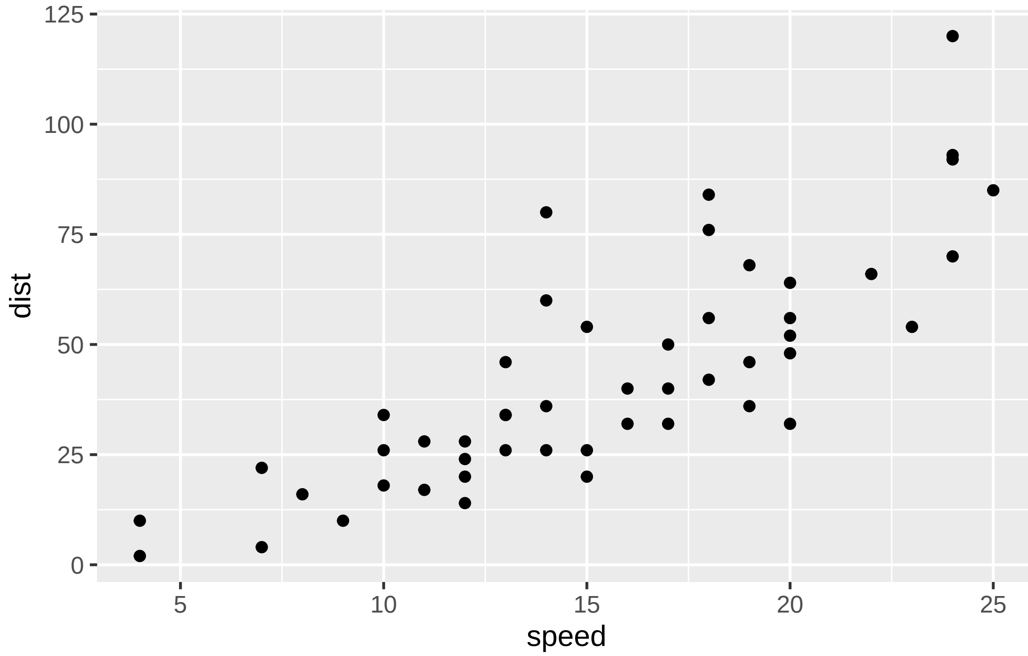
```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

Every ggplot has at least 3 layers:

- the **data** (a data.frame of the stuff we want to plot)
- the **aes**thetics(how data maps to plot)
- the **geom** layer (how you want the plot drawn, e.g. points, lines, etc.) The main function in the **ggplot2** package is called `ggplot()`

```
ggplot(cars) +
  aes(x=speed ,y=dist ) +
  geom_point()
```

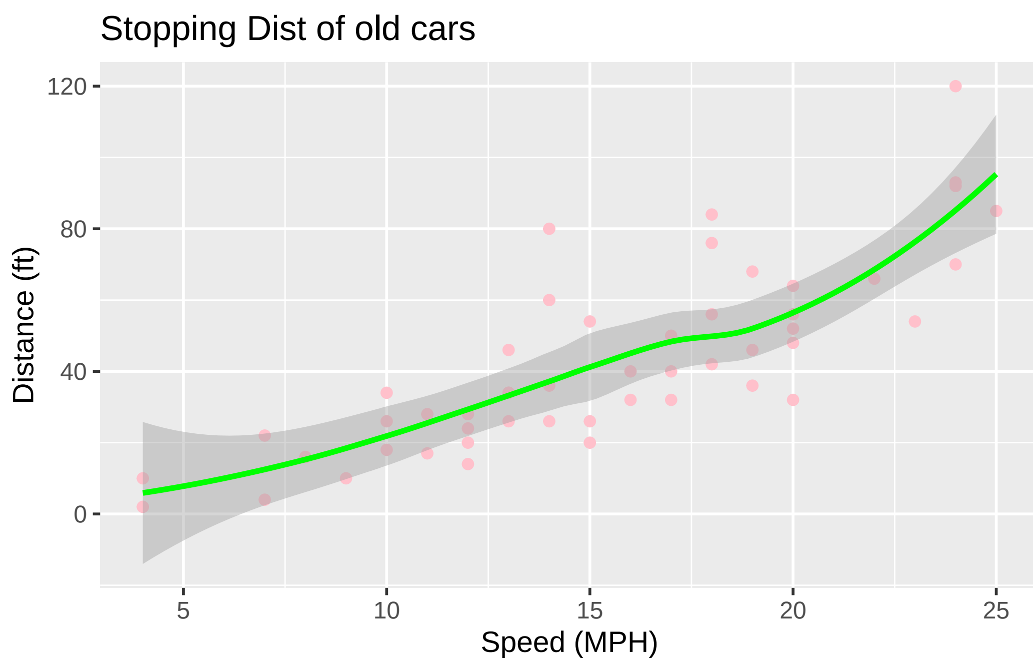


Add some custom features

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +
  aes(x=speed ,y=dist) +
  geom_point(color = "pink") +
  geom_smooth(color = "green") +
  labs(title = "Stopping Dist of old cars",
       x = "Speed (MPH)",
       y = "Distance (ft)")
```

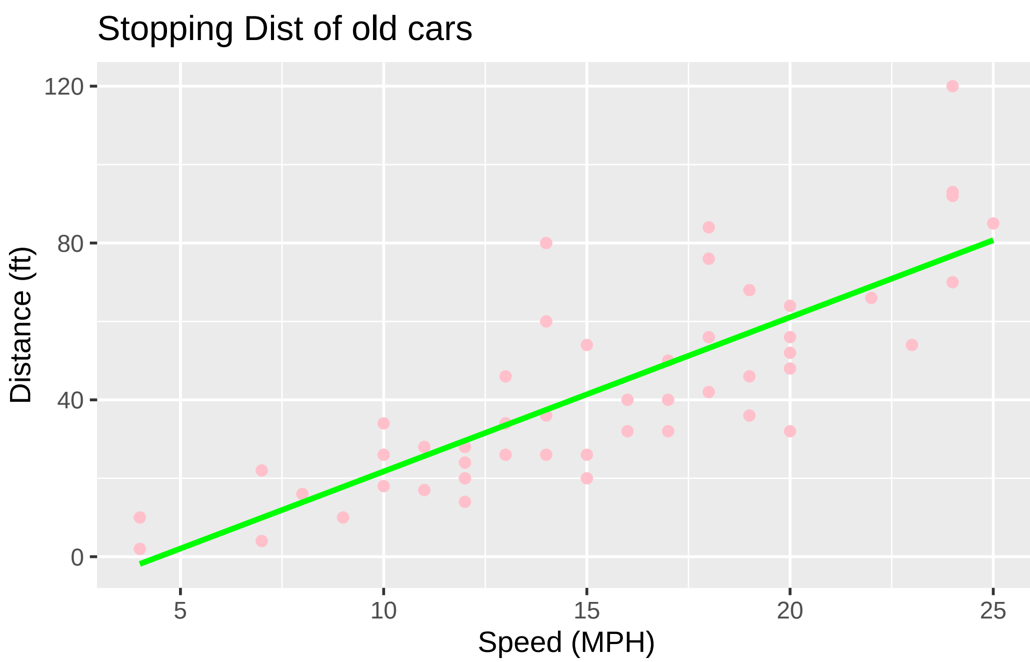
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Q. Can you make the `geom_smooth` function to produce a linear straight line fit to the data and turn off the "gray" error region?

```
ggplot(cars) +
  aes(x=speed ,y=dist) +
  geom_point(color = "pink") +
  geom_smooth(method = lm, se = FALSE, color = "green") +
  labs(title = "Stopping Dist of old cars",
       x = "Speed (MPH)",
       y = "Distance (ft)")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Figures

Import data to plot.

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
sum(genes$State == "up")
```

```
[1] 127
```

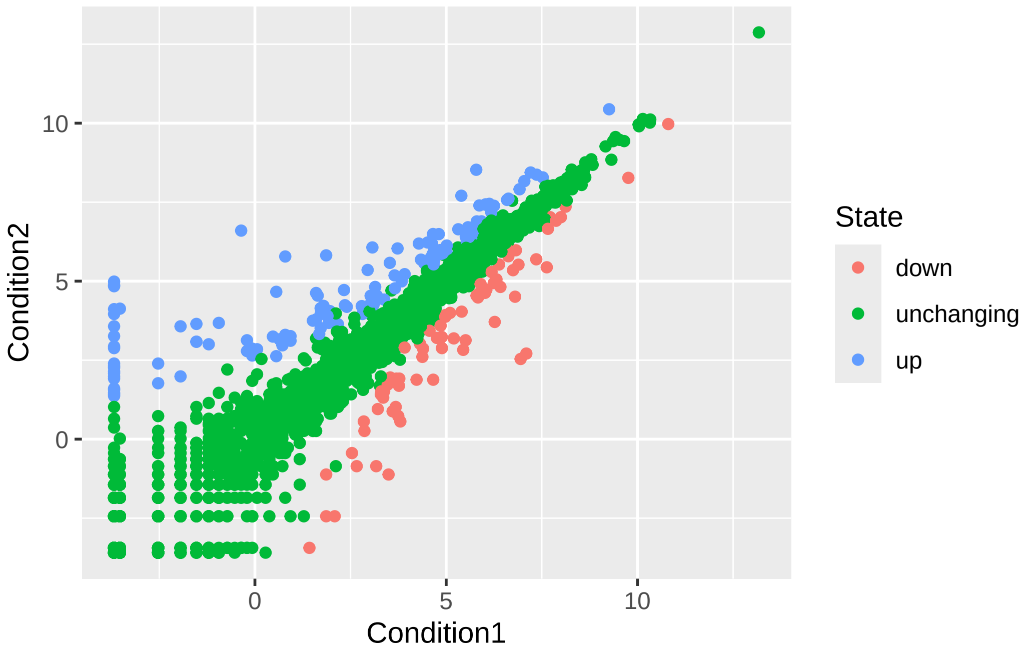
A new useful function in this context is the table() function:

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

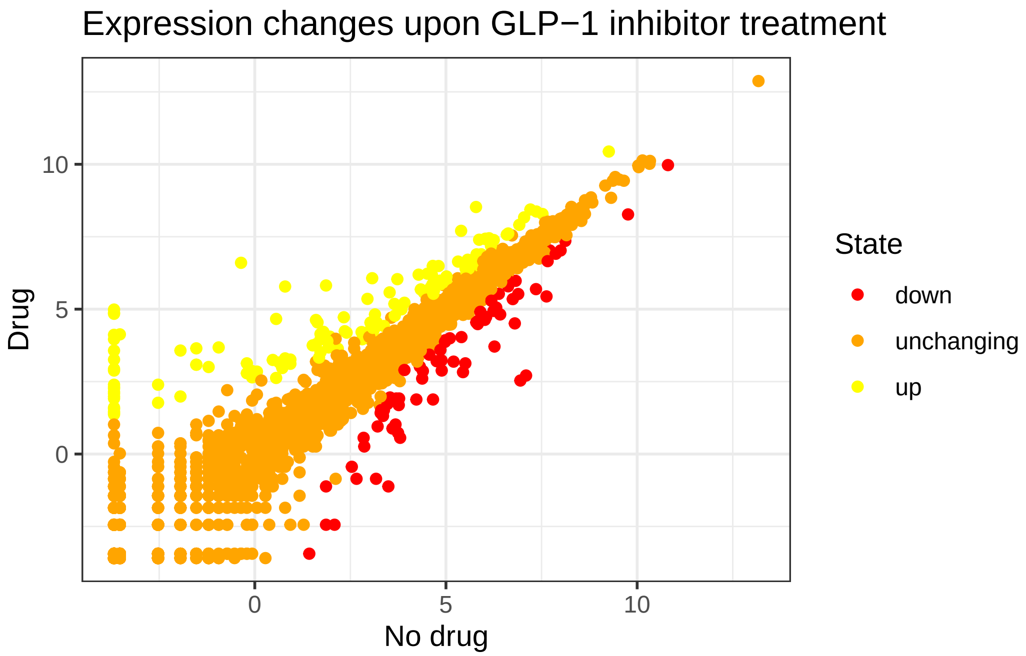
My first plot attempt

```
ggplot(genes)+
  aes(Condition1, Condition2, col=State) +
  geom_point()
```



Modified plot

```
ggplot(genes)+
  aes(Condition1, Condition2, col=State) +
  geom_point()+
  scale_color_manual(values=c("red", "orange", "yellow"))+
  theme_bw()+
  labs(x="No drug", y= "Drug",
       title = "Expression changes upon GLP-1 inhibitor treatment")
```



Going further

Here we read the famous gapminder dataset:

```r
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries(i.e rows) are in this dataset?

```r
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different "country" entries are in this dataset?

```r
length(table(gapminder$country))
```

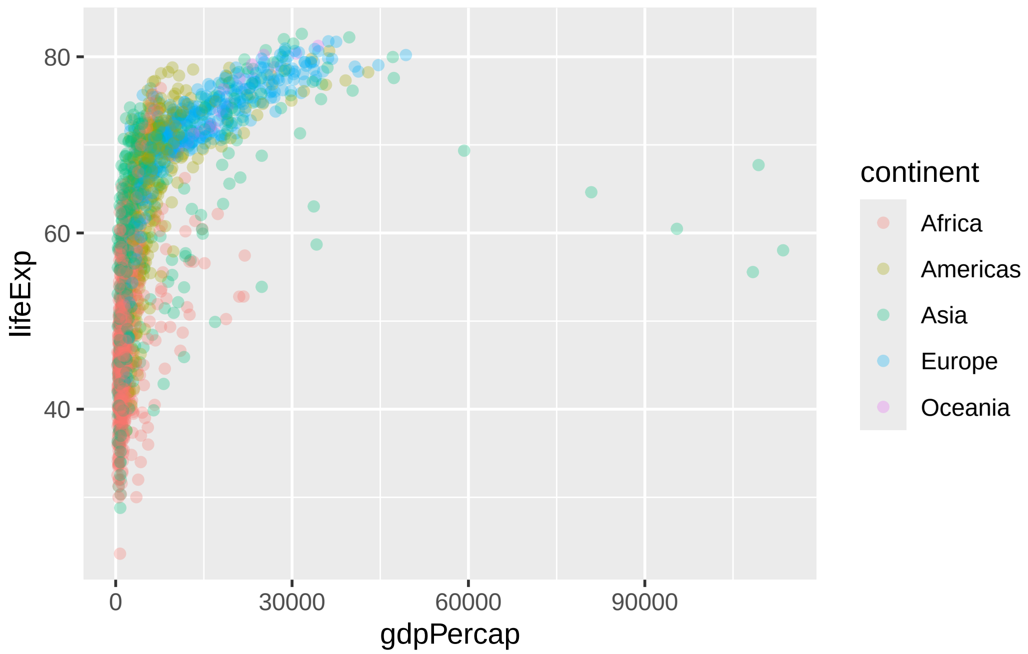
```
[1] 142
```

```r
#OR
length(unique(gapminder$country))
```

```
[1] 142
```

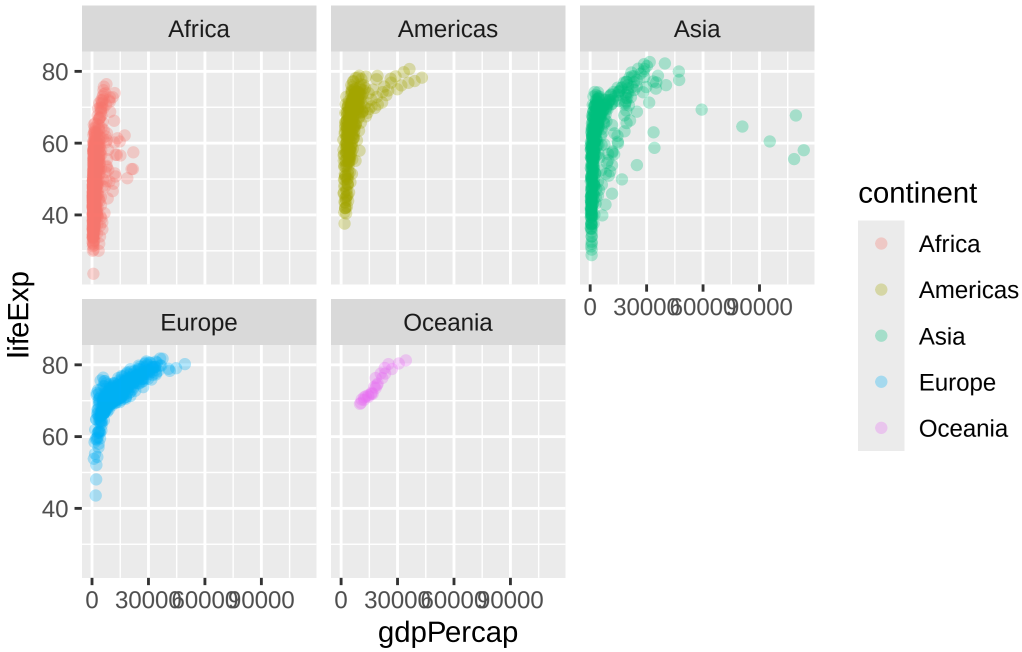
Let's make our first plot of the entire dataset: plot of "gdpPercap" vs" lifeExp" colored by "continent":

```r
p <- ggplot(gapminder)+
  aes(gdpPercap, lifeExp, col=continent)+
  geom_point(alpha = 0.3)
p
```



I can add more layers to p, the base plot.

```
p +
  facet_wrap(~continent)
```



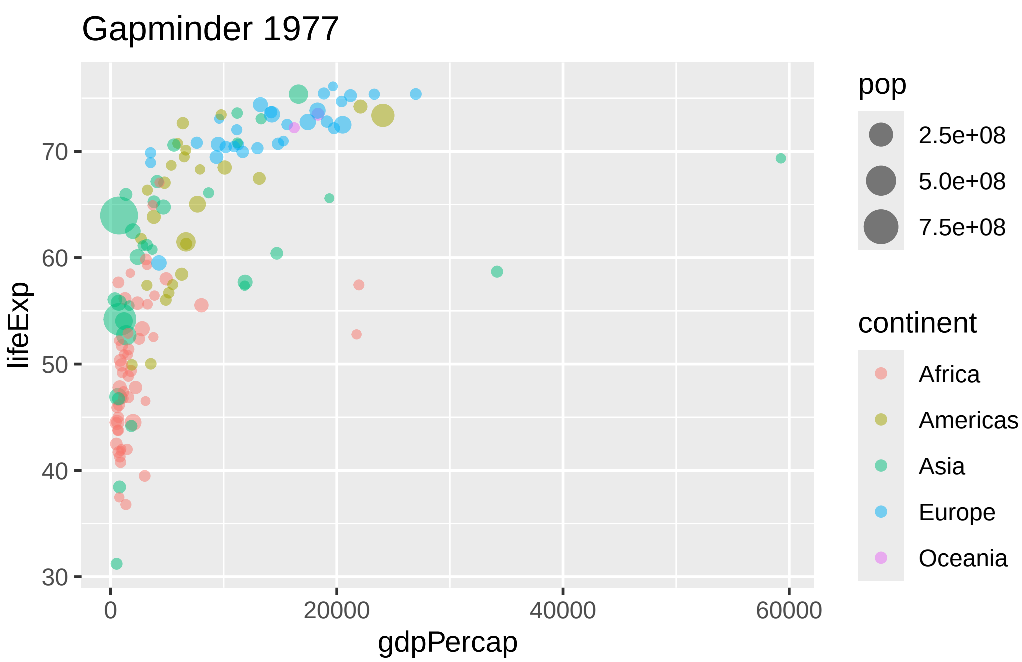
Make a plot for years 1977 and 2007 only(not all years in the dataset).

> Q. First use the **dplyr** package and the `filter()` function from that package to extract the year 2007.

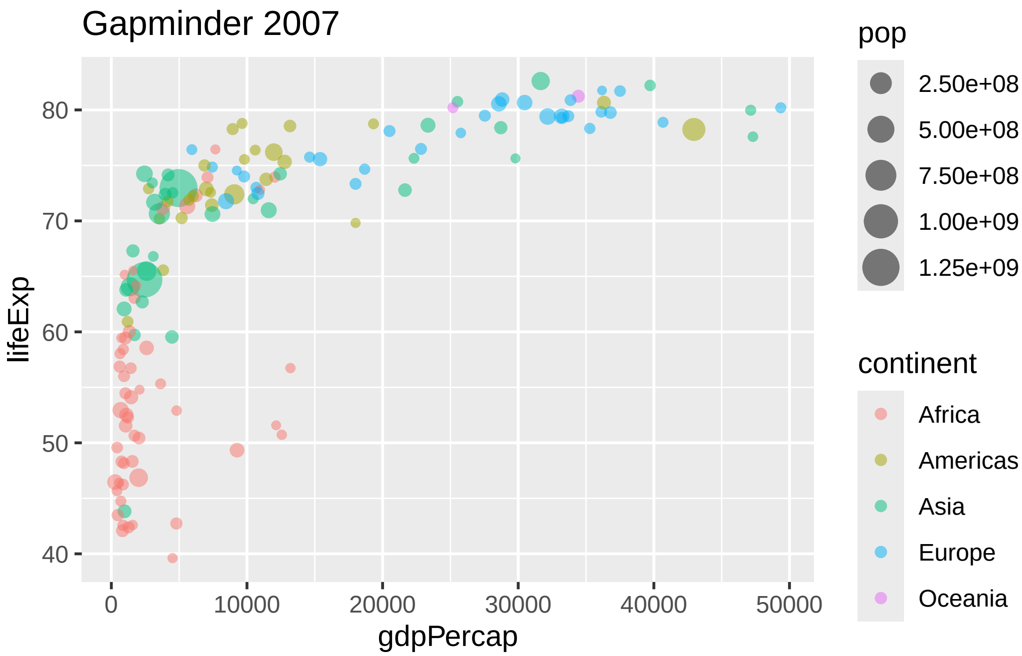
```
library(dplyr)
library(ggplot2)
```

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year == 1977)
```

```
ggplot(g77)+
  aes(gdpPercap, lifeExp, col=continent, size=pop)+
  geom_point(alpha = 0.5)+
  labs(title = "Gapminder 1977")
```



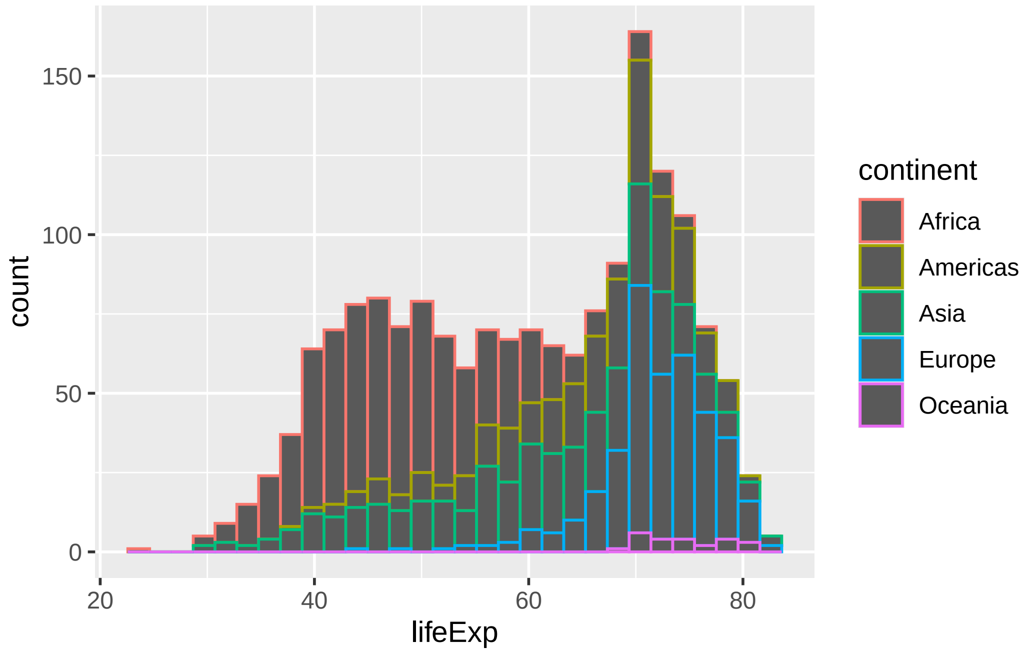

```r
ggplot(g07)+
  aes(gdpPercap, lifeExp, col=continent, size=pop)+
  geom_point(alpha = 0.5)+
  labs(title = "Gapminder 2007")
```



Q. Make a histogram of lifeExp colored by continent(use `fill=continent` or `col=continent`) Q. Make a histogram of lifeExp faceted by continent

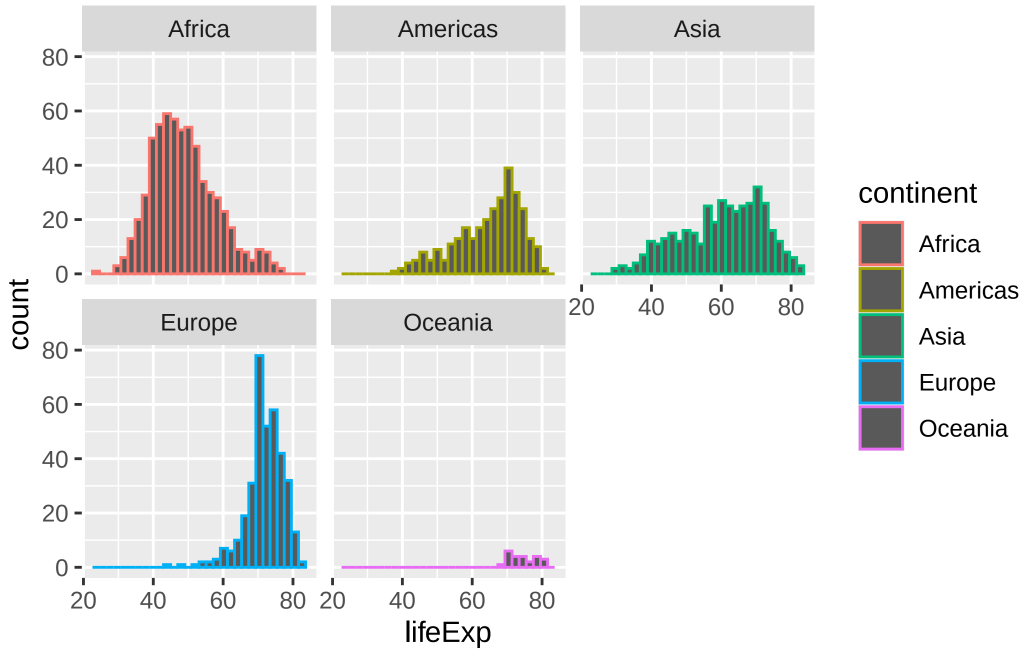
```
ggplot(gapminder)+
  aes(lifeExp, col=continent)+
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(gapminder)+
  aes(lifeExp, col=continent)+
  geom_histogram() +
  facet_wrap (~continent)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```


Africa
Americas
Asia
Europe
Oceania
count
lifeExp
0
20
40
60
80
continent
Africa
Americas
Asia
Europe
Oceania